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ARTICLE

# The Footbonaut as a new football-specific skills test: reproducibility and age-related differences in highly trained youth players

Adam Beavan[a,b,c], Job Fransen [c], Jan Spielmann[d], Jan Mayer[d], Sabrina Skorski[a] and Tim Meyer[a]

[a]Institute of Sports and Preventive Medicine, Saarland University, Saarbrücken, Germany; [b]Think Tank, German Football Association (Deutscher Fußball-Bund; DFB), Frankfurt, Germany; [c]Sport and Exercise Discipline Group, Faculty of Health, University of Technology, Sydney, Australia; [d]Performance Diagnostics, TSG 1899 Hoffenheim Fußball-Spielbetriebs GmbH, Zuzenhausen, Germany

**ABSTRACT**

**Purpose:** In sport, assessments are routinely administered to give an indication of performance. Assessing a skill requires external factors to dictate how and when the action is performed, highlighting the need for skill assessments to closely replicate the perception-action couplings experienced in football game play. Therefore, this study investigated if the Footbonaut is a valid and reliable football-specific skill assessment tool. **Methods:** Footbonaut performance scores from 152 male players from U12 to U23 representing a professional German Bundesliga club during the 2016/2017 season were analysed. **Results:** Pearson correlations ($r$) and coefficient of variation (CV) for the correct number of passes in a target (CV = 7.5–11.1; $r = 0.48$; $p < 0.001$), the speed at which they completed each trial (CV = 2.6–5.1; $r = 0.70$; $p < 0.001$), and a computer-generated point score (CV = 7.4–12.3; $r = 0.77$; $p < 0.001$) demonstrated acceptable test–retest reliability. Moreover, a MANOVA revealed a strong multivariate effect of age group on speed and accuracy combined ($F = 7.80$, $p < 0.001$, ES = 0.28), demonstrating the Footbonaut's construct validity. **Conclusion:** The results in this study demonstrated that the Footbonaut is a valid and reliable assessment of football-specific skill.

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**KEYWORDS**
Passing; perceptual-cognitive skill; reliability; validity; representative design

## Introduction

Football (Association Football) is a dynamic sport that requires players to utilise a highly automatic yet complex cohesion of physical fitness, football-specific motor skill, and perceptual-cognitive skill (Ali 2011). Many attempts have been made to capture football performance as a whole through a battery of laboratory and on-field assessments. When specifically assessing skill performance, it has been common practice to separate one aspect of the game such as dribbling performance (i.e., dribbling a ball around a pre-set course marked with cones) (Aquino et al. 2016) or ball control (i.e., playing six passes alternately against two opposing impact walls with at least two ball contacts) (Leyhr et al. 2018). Many talent identification programs use unique combinations of isolated tests to evaluate youth players' football performance, such as the Ghent Youth Soccer Project (Vaeyens et al. 2006). The combined results of the assessment batteries are routinely used for talent selection purposes, to compare changes in performance to normal baseline measures or to examine the effectiveness of a training regime (Serpiello et al. 2017). Although there are closed skills in football such as free kicks, football is predominately a dynamic environment in which players find functional movement solutions to movement problems that occur in constantly changing environments. Isolating a single aspect of the game should be considered a measure of technical ability rather than a demonstration of skill (Strand and Wilson 1993).

The design of a new assessment of football-specific skill is challenging due to the complex relationship of many physical and perceptual determinants of playing ability (Williams 2000). In order to assess football-specific skill, the test must allow for external factors to dictate how and when the actions are performed, in a dynamic and unpredictable context (Krause et al. 2018). However, a major barrier in the design of any skill assessment that aims to replicate the requirements of real-world environments is to provide the player with the optimal amount of uncertainty that mirrors game conditions, yet includes enough precision in the protocol to allow for test reliability (Ali 2011).

The most common performance assessment of skill is passing ability. Passing is an essential aspect of football, as it has been previously reported that a short pass preceded 47% of goals scored by direct shots in the 2006 FIFA World Cup (Sajadi and Rahnama 2007). Many researchers have devised tests to examine passing ability such as the Loughborough Soccer Passing Test (LSPT) (Ali et al. 2007) or the physical and technical test (PT-test) (Rostgaard et al. 2008). Although these assessments have obvious merit to examine technical abilities, the LSPT was reported to not be representative of in-game passing performance (Serpiello et al. 2017). This may be the result of its uni-dimensional task constraints with no manipulation of task performance.

Thus, there is still a paucity of literature related to football-specific skill assessments that more closely mirror the constraints players encounter during in-game performance. The Footbonaut (CGoal GmbH, Berlin, Germany) is a relatively new training and

**CONTACT** Adam Beavan adam.beavan@uni-saarland.de Institute of Sports and Preventive Medicine, Saarland University, Saarbrücken, Germany
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assessment tool developed in 2013. The unique design of the Footbonaut allows for the serial coupling between perception and action with the addition of unpredictability allowing for a better measurement of skill than that of previous assessments. A player in the Footbonaut can undergo an individualised training session to improve their skills, such as training their weaker foot's passing performance and first touch. Moreover, each player can be assessed on a standardised protocol to allow staff to monitor an individual's performance throughout the season, analyse between or within-group comparisons, and create age group norms for talent identification purposes. Performance in the Footbonaut is measured by a combination of the accuracy of the pass, and the speed at which the player carries out this movement in each trial. See Figure 1 for a schematic drawing of the Footbonaut.

Practitioners and scientists alike depend on the results of an assessment task to be a meaningful and a true representation of performance. Thus, discriminant validity and reproducibility of the Footbonaut are a necessary prerequisite for its further implementation as an assessment tool. The aims of the present study were twofold: (1) to assess the test–retest reliability of the Footbonaut and (2) to assess its discriminant validity by investigating the influence of age and football experience on test performance. It was hypothesised that the Footbonaut is a reliable assessment instrument and that older players would perform better in the Footbonaut compared to younger players.

## Materials and methods

### Participants

Retrospective data on male youth football players in eight age groups: U12 ($n$ = 18), U13 ($n$ = 20), U14 ($n$ = 18), U15 ($n$ = 20), U16 ($n$ = 20), U17 ($n$ = 8), U19 ($n$ = 25), and U23 ($n$ = 23) representing a professional German Bundesliga club during the pre-season/in-season of 2016/2017 were analysed. Each age group played at the highest level for their region and age. All participants (and parents for players under 18 years of age) signed contracts confirming that data arising as a condition of player monitoring over the course of the competitive season can be used for scientific purposes (Winter and Maughan 2009). The Faculty of Empirical Human Sciences and Economics ethics committee of Saarland University reviewed and approved this study.

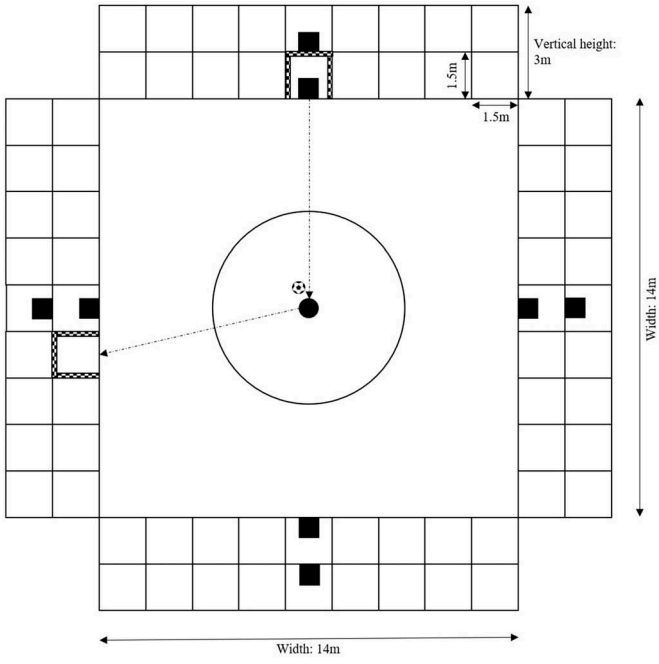


Figure 1. A schematic drawing on the Footbonaut.

### Skill test design

The Footbonaut is designed to be a realistic training tool that mirrors the dynamic elements of football gameplay by incorporating both physical (dribbling, passing, and shooting) and perceptual-cognitive (information-processing from auditory and visual cues) components. The Footbonaut consists of a - 14 × 14 m artificial turf surface (Morton Extrusionstechnik GmbH, Absteinach, Germany) and is surrounded by four walls. Together, these walls consist of 72 positioned square panels (64 target gates and 8 ball dispenser gates each 1.5 × 1.5 m, with two horizontal rows of gates per wall). Each gate is equipped with light barriers around the perimeter and light-emitting diodes to measure the ball exiting and entering gates.

### Procedure

Each age group was assessed on separate days but followed a standardised procedure. Players did not conduct any physical training the morning of the assessment. Players entered the Footbonaut and started the session in the centre zone. Before commencing the test, players had a practice session consisting of 10 trials to familiarise themselves with the equipment. Players received the instruction to 'perform the assessment as fast and accurate as possible'. Each session consisted of a standardised combination of 32 consecutive balls (see Table 1). A countdown of four seconds preceded the commencement of the test. Within each trial, an auditory and visual cue informed the participant which specific gate the ball would be dispensed from, and a second auditory and visual cue immediately identified the location of the target gate the player had to pass the ball into. The visual cue for the ball dispenser gate was a red light illuminated along the perimeter of the panel. The visual cue for the target was a green light illuminated around the perimeter of the specific target gate. If the player did not pass the ball into the target after two seconds of the ball being dispensed, the target light would change colour to blue. After five seconds the light would change colour to white, and the trial outcome was considered to have resulted in '*no goal*'. After the ball entered through a gate, the player had to swiftly return to the middle of the test zone to receive the next ball. The athletes performed two sessions on the same day separated by a 15-min resting period. No verbal encouragement during the session was given, nor were players told how many balls remained at any point. Upon completion of each session, players were provided with knowledge of results (accuracy and speed) (see Table 1).

**Table 1.** The parameters and performance variables defined for each session in the Footbonaut.

| Parameter | Value | Description |
|---|---|---|
| Balls | 32 | Number of balls dispensed |
| Ball dispenser power | 50 km/h | Speed of the ball |
| Vertical angle | 2° | Angle of inclination of ball dispenser |
| Shot delay | 800 ms | Delay between when the auditory cue and the ball dispensed |
| Gates used | 360° | 360°/360° gates used |
| **Variables** | | |
| Accuracy | Percentage (%) | Correct number of passes through the gates |
| Speed | Seconds (s) | Time the ball is dispensed until it enters the gate |
| Points | Arbitrary unit (Au) | An algorithm comprised of accuracy, speed and location of gates in reference to the ball dispense |

*Note*: Only the bottom row gates and ball dispensers were used.

### *Statistical analyses*

Kolmogorov-Smirnov normality tests alongside visual inspections of data using boxplots and histograms were used to examine if the speed, accuracy and points from both sessions followed a normal distribution in the entire sample. Collectively, it was deemed that these variables were normally distributed and therefore warranted the use of parametric statistics.

Pearson's correlation ($r$) and the coefficient of variation (CV) were used to assess test–retest reliability for speed, accuracy, and points for the entire sample. The interpretation of the Pearson's correlation coefficients followed the guidelines by Evans (1996), where correlations between 0.00–0.19, 0.20–0.39, 0.40–0.59, 0.60–0.79 and 0.80–100 are considered very weak, weak, moderate, strong and very strong respectively. Furthermore, discriminant validity of the Footbonaut was assessed by examining changes in speed, accuracy, and points based on age groups. Therefore, a multivariate analysis of variance () was used to assess the effect of age on Footbonaut performance using the age group that the players belonged to (U12–U23) as a fixed factor and the speed and accuracy variables generated by the Footbonaut as dependent variables. As the variable 'points' are calculated using an algorithm that includes speed and accuracy components among other factors such as the spin of the ball and the location of the ball release in regard to target location, a separate Analysis of Variance (ANOVA) with points as the dependent variable and age group as a fixed factor was used to analyse between group differences in points. Bonferroni corrections were used to investigate multiple comparisons between age groups and partial eta squared effect sizes (ES) were used throughout to investigate the magnitude of any observed effects using Cohen (1988) guidelines for interpreting ES: 0.01–0.06 = small effect, 0.06–0.14 = moderate effect and >0.14 = large effect.

Finally, as the Footbonaut is a commercially available tool and the algorithm related to the calculation of the Footbonaut point score is not readily available, a linear regression with points as the dependent and speed and accuracy as independent variables was used to analyse the variance in the points that can be attributed to a player's accuracy and speed. For all analyses, the significance level was set at $p < 0.05$.

## Results

Pearson correlations ($r$) revealed moderate to strong relationships between the test and retest scores for the entire sample, and the dispersion measured through CV values was adequate and comparable to previous tests of football skill for all age groups (Ali et al. 2007) (see Figure 2). More specifically, accuracy ($CV_{mean} = 8.68$; $r = 0.48$; $p < 0.001$), speed ($CV_{mean} = 3.60$; $r = 0.70$; $p < 0.001$), and points ($CV_{mean} = 8.93$; $r = 0.77$; $p < 0.001$) demonstrated acceptable test–retest reliability.

A MANOVA revealed a strong multivariate effect of age group on speed and accuracy combined ($F = 7.80$, $p < 0.001$, ES = 0.28). Further univariate analysis revealed strong age group effects on speed ($F = 14.88$, $p < 0.001$, ES = 0.42) and accuracy ($F = 3.77$, $p = 0.001$, ES = 0.16) components of players' Footbonaut performance. More specifically, U19 and U23 players were more accurate than U12 players, and speed decreased with increasing age in the youngest cohort of players (U12–U14), but no more differences in speed were observed between players in the U15–U23 categories (see Table 2). Additionally, the ANOVA revealed strong age group effects on points ($F = 17.96$, $p < 0.001$, ES = 0.47) in which a similar trend was observed for the effect of age group on speed.

Finally, a multiple regression analysis showed that 88% of the variance ($F = 539.10$, $p < 0.001$) was explained by speed and accuracy components. Both speed and accuracy were significant predictors of points, but speed seemed to be a bigger contributor (beta = −0.75) than accuracy (beta = 0.38). The regression equation derived from this multiple regression is: 142,902 + 469.73*Accuracy(%) − 49,981*Speed(s). Confidence intervals for the constant and these coefficients are 130678.24–155126.51, 396.80–542.66, and −53922.95 to −46039.39 respectively.

## Discussion

The main findings of this study were that the Footbonaut is a reliable tool across all age groups and some discriminant validity was observed through performance differences between younger and older age cohorts in this study. The results confirmed the hypothesis that older players performed better in the Footbonaut compared to their relatively younger counterparts; however, the age performance differences were less pronounced in accuracy in comparison to speed and points.

The Footbonaut is a relatively new tool allowing athletes to be assessed in a standardised and representative football environment that more closely replicates the serial coupling of perception and action experienced during football match play. Results revealed moderate to strong relationships between test and retest scores for the entire sample of players. According to previous research in football, a strong relationship exists between accumulated sport specific practice and better performances on

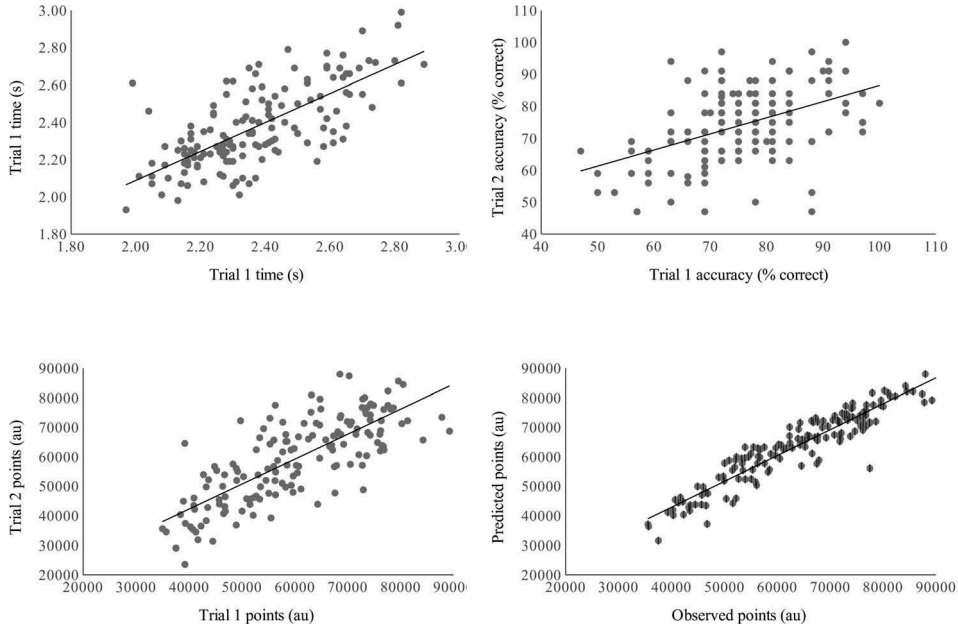


**Figure 2.** (A–C) Scatterplots demonstrating test–retest reliability for speed, accuracy and points components of U12–U23 players' Footbonaut performance. (D) Predicted versus observed scores for a multiple regression analysis in which points are predicted by speed and accuracy of U12–U23 football players' Footbonaut performance.

**Table 2.** The performance variables mean (*SD*), *F*-values, significance levels and partial eta squared effect sizes (ES) between age groups for best speed, best score and best points out of two trials. Furthermore, the reliability measures (coefficient of variation ± 95% CI) from each age group.

| | U12 (*n* = 18) | U13 (*n* = 20) | U14 (*n* = 18) | U15 (*n* = 20) | U16 (*n* = 20) | U17 (*n* = 8) | U19 (*n* = 25) | U23 (*n* = 23) | *F* | ES |
|---|---|---|---|---|---|---|---|---|---|---|
| **Variables** | | | | | | | | | | |
| Height (cm) | Unknown | 156.9 (7.0) | 161.6 (8.3) | 170.6 (7.8) | 176.0 (6.6) | 176.5 (5.7) | 178.3 (7.4) | 184.1 (6.5) | | |
| Weight (kg) | Unknown | 42.9 (4.3) | 50.3 (9.8) | 60.8 (8.5) | 65.6 (7.1) | 70.6 (9.4) | 71.4 (8.3) | 78.9 (6.8) | | |
| **Performance variables** | | | | | | | | | | |
| Speed (s) | 2.58 (0.16) | 2.48 (0.17) | 2.36 (0.12)[#] | 2.32 (0.18)[#,$] | 2.26 (0.16)[#,$] | 2.14 (0.09)[#,$,@] | 2.24 (0.16)[#,$] | 2.21 (0.14)[#,$] | 14.881** | 0.42 |
| Accuracy (% correct) | 71.0 (10.0) | 79.7 (11.6) | 75.3 (11.0) | 78.3 (9.6) | 76.0 (8.8) | 82.1 (11.1) | 81.9 (10.6)[#] | 84.7 (7.6)[#] | 3.773** | 0.16 |
| **ANOVA** | | | | | | | | | | |
| Points (Au) | 46,023.1 (6340.2) | 54,566.5 (9322.2) | 57,938.3 (9888.4)[#] | 65,639.6 (11,106.9)[#,$] | 69,266.8 (11,675.6)[#,$,@] | 74,823.6 (5304.9)[#,$,@] | 68,541.0 (11,082.3)[#,$,@] | 72,814.0 (8901.0)[#,$,@] | 17.959** | 0.47 |
| **Coefficient of variation (%)** | | | | | | | | | | |
| Speed (s) | 4.5 (3.2–5.8) | 2.6 (1.5–3.7) | 3.9 (2.4–5.5) | 2.5 (1.3–3.7) | 4.5 (2.6–6.3) | 5.1 (2.0–8.1) | 3.1 (2.0–4.2) | 3.7 (2.1–5.3) | | |
| Accuracy (% correct) | 8.3 (5.6–10.9) | 9.6 (5.6–13.5) | 10.1 (5.4–14.9) | 7.6 (4.9–10.3) | 7.5 (5.0–10.0) | 11.1 (4.1–18.0) | 7.7 (5.3–10.2) | 7.5 (3.8–11.1) | | |
| Points (Au) | 9.8 (6.0–13.6) | 9.3 (6.4–12.2) | 7.9 (4.5–11.3) | 8.1 (5.4–10.8) | 9.8 (5.3–14.2) | 12.3 (4.7–19.9) | 7.4 (5.3–9.4) | 9.2 (6.0–12.3) | | |

*Notes*: ** = $p < 0.01$, # = different from U12, $ = different from U13, and @ = different from U14.

assessments of technical and tactical skills (Hendry et al. 2018). Therefore, it was hypothesised that there would be a positive linear relationship between age and Footbonaut performance. Although each performance variable demonstrated trends of improvement with incremental increases in age, there were no other significant differences observed between players in the U15–U23 categories. Thus, the discriminant validity of the Footbonaut disappears when groups become more homogeneous after the age of 15 years.

The observed plateau of skill differences after the age of 15 reported in this study are in line with recent research on the development of skill in youth football players. A recent study that analysed three years of athletic development of highly talented young football players in the German Soccer Association's talent identification program reported that substantial improvements in participants' performance in motor skill assessments occurred from U12 to U15 (Leyhr et al. 2018). The progression was non-linear, where players' motor performance increased considerably from U12 to U13 and became less prominent in subsequent years (i.e., from U13 to U14 and U14 to U15). The results of this study are also in line with Huijgen (2013) reporting that players' dribbling performance improved largely between U13 and U15 followed by a phase of development plateau after the U15 age group.

Another recent study noted that it may be between 15 and 17 years old that skill refinement is significant during youth

development, and where perceptual-cognitive skills begin to become more apparent (Hendry et al. 2018). Accordingly, a common theme shared amongst the current study and previously used testing protocols is that they may not have been challenging or sensitive enough to measure any further developments in perceptual-cognitive skills after the age of 15. If technical skills begin to plateau after the age of 15, then alternative approaches may be to develop more age-appropriate test protocols that place greater demands on the perceptual-cognitive component in older cohorts. For example, Fransen et al. (2017) manipulated the perceptual-cognitive component in a soccer-specific dribbling task by restricting visual feedback with stroboscopic glasses. Similar concepts may strengthen the discriminant validity of the Footbonaut in older age groups. However, any adjustment to the standard protocol used in this test would require further evaluation (McDermott et al. 2015). Additionally, as many of the findings in previous research are attributed to maturational changes in individuals during early adolescence (Malina et al. 2007; Huijgen et al. 2010; Leyhr et al. 2018), future research should also consider measuring the influence that physical and cognitive maturation has on performance in the Footbonaut.

Despite the numerous advantages of using the Footbonaut, a number of limitations should be considered when interpreting the results of the present study. First, the Footbonaut is an expensive tool (estimated USD $3.5 million) that is not widely available or accessible for the majority of football clubs. Thus, the development of a more affordable skills assessment test that mirrors the reciprocal relationship between domain-specific and perceptual-cognitive skills of the Footbonaut has value. Second, unlike the LSPT, the Footbonaut cannot distinguish between gross and fine motor performance in the session, and where the errors lie in a slower test completion time. Future research should incorporate the analysis of video footage within the Footbonaut to measure performance aspects like the 'first touch'. Lastly, it is not clear what performance variables are most beneficial to performance in the Footbonaut. For instance, it is not evident whether older groups are physically better with similar perceptual-cognitive abilities as the younger groups, or that older players are superior at planning their executions of movements from better perceptual-cognitive skill. To overcome this limitation, future research should explore the cofounding effects of perceptual-cognitive skills (i.e., visual scanning behaviour prior to receiving the ball (McGuckian et al. 2018)) and physical fitness (i.e., speed and agility) on performance in the Footbonaut. Thus, the addition of isolated tests of domain-specific skills may compliment the Footbonaut within a battery of tests to provide further insight into what strengths and weakness' in performance a player yields.

## Conclusion

The results demonstrated that the Footbonaut is a reliable and valid tool to differentiate between age groups in youth male football players but has stronger discriminant validity in cohorts below U15. More research is needed to investigate which parameters of performance can be added to the Footbonaut to improve the discriminant validity between older age groups in youth football. Furthermore, practitioners should be aware of the limitations of using skill tests that isolate single aspects of the football such as measuring dribbling performance using a ball around pre-set cones, as such tests might not be representative of in-game performance. Therefore, representative assessments of football-specific skill should involve a combination of both physical and perceptual-cognitive components.

## Practical Implications

The Footbonaut is a tool with the potential to open new avenues for using representative research designs that can further understanding of the reciprocal nature of the perception-action coupling in football. The findings of this study are valuable for practitioners who want to design representative tasks, as the Footbonaut can be replicated in the field by coupling passing actions with decision making and measuring speed and accuracy.

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## Disclosure statement

No potential conflict of interest was reported by the authors.

## ORCID

Job Fransen http://orcid.org/0000-0003-3355-1848

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